A Two-Step ADC With a Continuous-Time SAR-Based First Stage

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***Abstract*— This article presents a two-step analog-to-digital converter (ADC) that operates its first-stage successive approximation register (SAR) ADC in the continuous-time (CT) domain. It avoids the front-end sample-and-hold (S/H) circuit and its associated sampling noise. Hence, the proposed ADC allows the input capacitor size to be substantially reduced without incurring large sampling noise penalty. With input ac coupling, the first-stage CT-SAR can simultaneously perform input tracking and SAR quantization. Its conversion error is minimized by accelerating the SAR speed and providing redundancy. A floating-inverter-based (FIB) dynamic amplifier (DA) is used as the inter-stage amplifier and acts as a low-pass filter for the first-stage residue. To verify the proposed techniques, a 13-bit prototype ADC is built in 40-nm CMOS process. Its input capacitor is only 120 fF, which is over 20 times smaller than what would be needed in a classic Nyquist ADC with the S/H circuit. Operating at 2 MS/s, it achieves 72-dB signal-to-noise-and-distortion-ratio (SNDR) at the Nyquist rate while consuming only 25 μW of power and 0.01 mm^2 of area.**

***Index Terms*— Analog-to-digital converter (ADC), continuous-time SAR (CT-SAR), dynamic amplifier (DA), sampling noise reduction, successive approximation register (SAR)**

I. Introduction

A DISCRETE-TIME (DT) analog-to-digital converter (ADC) has a front-end sample-and-hold (S/H) circuit. Fig. 1(a) shows an example of a classic two-step successive approximation register (SAR) ADC with the front-end sampler. The benefit of having a front-end S/H circuit is that it converts a continuous-time (CT) input into a DT signal that stays unchanged between samples, which simplifies the following analog signal processing operations (e.g., quantization, subtraction, and amplification). Nevertheless, the use of the front-end S/H brings an unwanted sampling noise, which poses a fundamental SNR limit for the ADC. This sampling noise is typically suppressed passively by increasing the capacitor size [1]–[13]. For example, the total differential sampling capacitors need to be greater than 2.1 pF to achieve a sampling noise-limited SNR of 80 dB with a 2.5-V peak-to-peak differential signal swing, and it has to be quadrupled for every 1-bit increase in the resolution. Such a large input capacitor makes it difficult to design the S/H circuit and leads to increased ADC power and area. Moreover, as indicated in Fig. 1(a), a large input capacitor poses critical challenges for both the ADC buffer and the reference buffer. To meet the stringent linearity requirement of high-resolution ADCs, these buffers consume a significant amount of power, which can be comparable or even higher than the ADC itself. Thus, it is highly desirable to look for a way to break this tight tradeoff between the sampling noise and the capacitor size, so that a small input capacitor can be used but without incurring significant noise penalty.

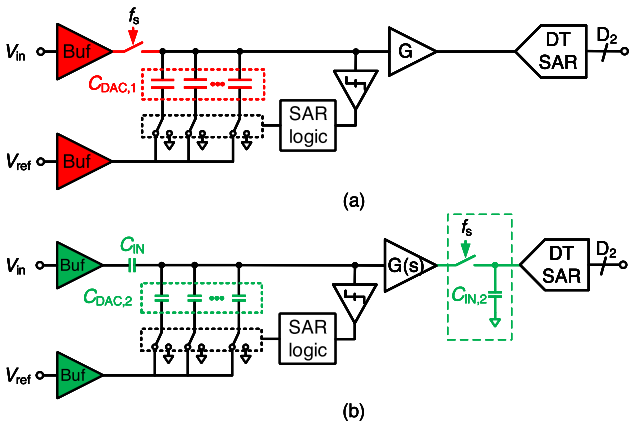


Fig. 1. (a) Conventional DT two-step SAR ADC. (b) Proposed CT-SAR-assisted two-step SAR ADC.

The core idea explored by this article is to remove the front-end S/H circuit and operate the first stage of the two-step ADC in the CT domain [14], [15]. Fig. 1(b) shows the basic architecture of the proposed ADC. In the first-stage CT-SAR, the in-band thermal noise due to the switch resistance is much smaller than kT/C, as its sampling-free CT operation prevents wide-band noise folding, which, otherwise, would be the dominant contributor to the sampling noise. The inter-stage amplifier also processes the first-stage conversion residue in CT and acts as a low-pass filter that suppresses the out-of-band thermal noise from the switch resistors. Consequently, the first-stage

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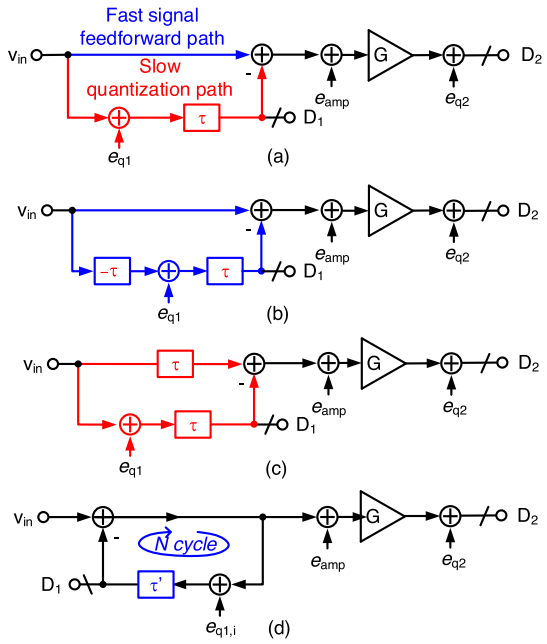


Fig. 2. Block diagrams of CT pipeline ADC with (a) delay mismatch, (b) negative delay added to the slow path, (c) positive delay added to the fast path, and (d) proposed CT-SAR with shortened delay and built-in redundancy.

capacitor size is not bounded by the kT/C limit and can be significantly reduced. Unlike the classic two-step ADC, the sampling operation is moved to the second stage, as shown in Fig. 1(b). Although the second stage still suffers from its own sampling noise, this noise is substantially suppressed by the inter-stage gain when input-referred, thus permitting the use of a small capacitor in the second stage too. Overall, the use of the S/H-free CT first-stage breaks the link between the sampling noise and the capacitor size, making it possible to design high-resolution Nyquist-rate ADCs with small capacitors.

Although removing the S/H circuit brings the benefit of suppressed sampling noise, it can cause the first-stage conversion residue to go out of bound. Fig. 2(a) shows a simplified model of the two-step ADC. D_i represents the digital output code of the ith stage. e_{qi} represents the quantization noise added of the ith stage. e_{amp} represents the noise of the inter-stage amplifier. The input signal V_{in} goes through two different paths in the first stage. In the upper fast path, the signal directly goes to the inter-stage amplifier. In the lower slow path, the signal goes through the sub-ADC, the sub-digital-to-analog-converter (DAC), and the analog subtractor. All of these analog signal processing steps introduce extra delays, which are lumped into a single block τ. In a classic two-step ADC with the S/H circuit, this path delay mismatch is not an issue because the sampled input does not change. However, for the S/H free CT first-stage, this delay mismatch can cause output signals from the two paths to be misaligned, resulting in a much larger conversion residue and potentially saturating the inter-stage amplifier and the second-stage ADC.

One way to address this delay mismatch problem is to insert a negative delay $(-\tau)$ block in the slow path to cancel the positive delay [14], as shown in Fig. 2(b). Even though a pure negative delay is non-causal and unpractical, a negative delay within the signal band can be realized by using an analog prediction filter. However, it requires power-hungry wide-band op-amps to provide a full-signal-band prediction. Moreover, to satisfy causality, the analog prediction filter introduces a positive delay for out-of-band high-frequency signals. This increases the delay mismatch between the two paths, causing the first-stage conversion residue to increase for out-of-band signals. Hence, to keep the first-stage conversion residue within the allowable range, the anti-aliasing filter needs to provide stronger attenuation for out-of-band signals, which makes its design more challenging. Another approach to tackle the delay mismatch issue is to insert a positive delay block in the fast path [15], as shown in Fig. 2(c). This positive delay can be realized by using an LC lattice filter. This approach is fully passive and power efficient. Nevertheless, since it relies on delay matching, it requires careful tuning to compensate for process and temperature variations. Furthermore, while it works well for high-speed gigahertz operations, it is not well suited for low-to-medium speed applications (e.g., sensors), as the required long delay could result in large LC values and excessive chip area. In addition, it is non-trivial to design the filter to avoid amplitude attenuation or phase modulation.

This article seeks to address the delay mismatch problem in a different way. In a conventional two-step ADC, the slow quantization path consists of a multi-bit flash, a multi-bit DAC, and an analog subtractor, whose total aggregated delay tends to be relatively large. By contrast, the proposed ADC uses a CT-SAR-based first stage, as shown in Fig. 2(d). Since each SAR cycle contains only a single-bit comparison, a single-bit DAC, and a built-in subtraction operation, its delay is much shorter. $e_{q1,i}$ represents the quantization noise added during the ith iteration. Even though a multi-bit quantization (e.g., 7-bit) requires multiple SAR cycles, their delays do not accumulate. This is because the CT-SAR operates on the CT input. Every SAR cycle sees its new instantaneous input. To further reduce the delay, the CT-SAR adopts asynchronous clocking [16] and dynamic logic [17], [18]. As the input moves during the CT-SAR operation, the prior SAR comparator decisions can be no longer correct. To tackle this problem, sufficient redundancy is prepared in each bit such that the CT-SAR can still tightly track the input. Once the CT-SAR finishes, the conversion residue is readily available. Thus, it allows the dynamic amplifier (DA) to be triggered immediately, minimizing the delay. With these techniques applied, the overall delay mismatch can be made very short. Hence, no prediction filter or LC lattice filter is needed in this article, leading to lower design complexity, as well as reduced chip area and power. In addition, different from prior CT pipelined ADCs [14], [15] that use input resistor and current-source DAC, this article uses input capacitor and capacitive DAC. Capacitors are noise-free and do not consume static current. As a result, the proposed ADC is more efficient from both noise and power perspectives.

To verify the proposed techniques, a 13-bit prototype ADC is built in 40-nm CMOS process. Its input capacitor is only 120 fF, which is over 20 times smaller than what would be needed in a classic DT two-step ADC. The inter-stage

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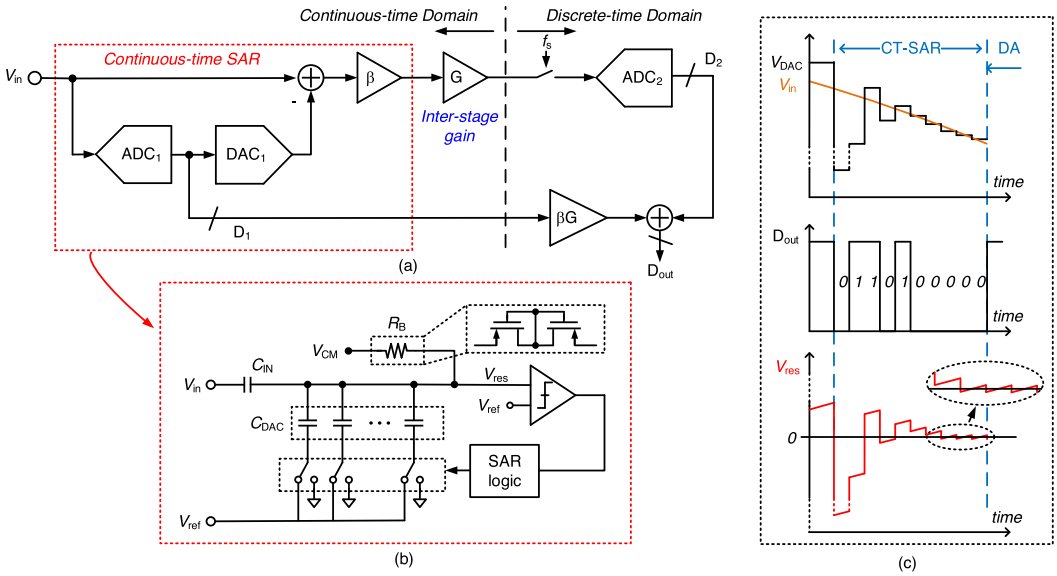


Fig. 3. Architectural block diagram of (a) proposed CT-SAR-assisted two-step SAR ADC, (b) first-stage CT-SAR ADC, and (c) example waveforms for key nodes.

amplifier in this article adopts a floating-inverter-based (FIB) DA topology. Comparing to classic static amplifiers, the proposed FIB DA has low-power and low-noise. Compared to the integrator-based DA [7], [19]–[21], it provides higher gain and stronger rejection to the input common-mode (CM) variations. Operating at 2 MS/s, the ADC achieves 72-dB signal-to-noise-and-distortion-ratio (SNDR) across the Nyquist band while consuming only 25 μW of power and 0.01 mm^2 of area.

This article is an extension of [22] and is organized as follows. Section II presents the operation principle of the proposed ADC. Section III presents the implementation details of the prototype ADC. The measurement results are discussed in Section IV. Finally, Section V draws the conclusion.

II. Proposed Two-Step ADC With First-Stage CT-SAR

A. Topology Overview

Fig. 3(a) shows the block diagram of the proposed CT-SAR assisted two-step ADC. Its first stage works in the CT domain, while the second stage works in the DT domain. The first stage CT-SAR performs a CT approximation of the input and produces a CT residue. The inter-stage amplifier amplifies the residue and also filters out the wide-band thermal noise due to its low-pass response. The sampler comes after the amplifier and converts the signal into the DT domain for the second-stage DT-SAR. As mentioned earlier, even though this sampler produces sampling noise, since it is after the inter-stage gain block, this noise is greatly attenuated when input-referred. The final ADC output is the weighted sum of digital outputs from the first-stage CT-SAR and the second-stage DT-SAR.

In a classic DT-SAR, a single-capacitor array is used for both input sampling and SAR DAC in a time-duplex manner. The benefits are less number of capacitors and no signal attenuation. However, this scheme only works for the DT-SAR with the S/H circuit. To realize the CT-SAR and ensure simultaneous input tracking and SAR conversion, the input capacitor and the SAR DAC have to be separated, as shown in Fig. 3(b). The cost is more capacitors and an attenuation factor of $\beta \equiv C_{\text{in}}/C_T$ from V_{in} to V_{res}, where C_T represents all the capacitance at V_{res}, including C_{in}, C_{DAC}, and the parasitic capacitance (including the input capacitance of the comparator and the amplifier). The signal attenuation results in an increased noise contribution from the inter-stage amplifier. Thanks to the reduced first-stage sampling noise, the inter-stage amplifier can produce more noise while keeping the total ADC input-referred noise unchanged. As a result, the increase in amplifier power is not significant. Overall, this cost is worthwhile as the sampling noise can be greatly suppressed and the capacitor size can be substantially reduced, which can lead to not only significant area savings for the ADC core itself but also potential power savings for the input and reference buffers. The detailed noise analysis is given in Appendix V.

As shown in Fig. 3(c), the CT-SAR performs a CT approximation for the input. The residue V_{res}, which is the difference between the input and the SAR DAC output, is updated during every SAR cycle. Since the comparator always sees the CT input, the delays from different SAR cycles do not accumulate, as long as the CT-SAR can accurately track V_{in}. The final CT-SAR conversion result D_1 corresponds to the instantaneous ADC input at the time of the LSB comparison, rather than the MSB comparison. Once the CT-SAR finishes, the residue V_{res} is readily available at the comparator input, and thus, the inter-stage amplifier can be triggered immediately. This helps to ensure the close match between V_{in} and D_1. Any extra delay can cause the increase of V_{res} due to the variation in $V_{\text{in}}(t)$.

Unlike prior CT pipeline works that use input resistive coupling and a current-source DAC (IDAC) [14], [15], this article uses input capacitive coupling and a capacitor DAC (CDAC). It eliminates the noise from the input resistor and the IDAC. It also removes the static power from the IDAC.

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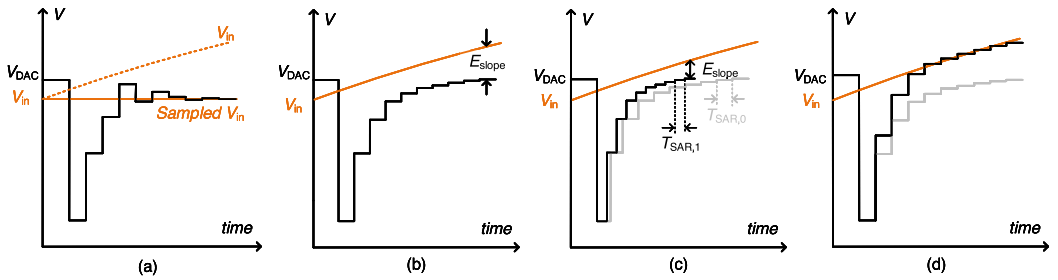


Fig. 4. Input and DAC output example waveform for (a) conventional DT-SAR ADC, (b) large E_{slope} for CT-SAR ADC by simply removing S/H circuits, (c) recovered residue voltage with accelerated SAR conversion, and (d) recovered residue voltage with built-in redundancy.

Yet, the tradeoff is that capacitive coupling blocks dc signals. A pseudo-resistor R_B is used to provide the dc bias for the comparator input [23], [24]. R_B and the capacitors form a high-pass filter that blocks the low-frequency input. The proposed capacitively coupled two-step ADC can be used for a wide range of applications where the information does not reside at dc, such as audio, biological, and communication signals by setting the proper passband frequencies.

B. CT-SAR Conversion Error and Mitigation

Unlike a DT-SAR, the input varies with time during one CT-SAR conversion process, which can cause extra conversion error. To illustrate this key difference, Fig. 4(a) shows the case for a conventional DT-SAR. The DAC output is compared to a sampled input, which does not change during the SAR conversion. As a result, the sampled input can be precisely converted regardless of how different the real-time input deviates from the sampled one. The conversion error is limited only by the quantization step, if we ignore other circuit non-idealities (e.g., capacitor mismatch, comparator noise, DAC settling error, etc). The S/H circuit essentially isolates the CT input from the SAR conversion process.

By contrast, without the S/H, the CT-SAR is fully exposed to the CT input, whose variation can cause a large error as the SAR conversion process no longer has a consistent convergence target. Fig. 4(b) shows an example that a rising input signal happens to be slightly below the decision threshold of the MSB comparison. The comparator outputs a '0', and directs the following binary searches to be below the MSB decision threshold. However, the input actually rises above the MSB threshold as time goes by. As a result, the DAC output would fail to track the time-varying input, leading to a large conversion error. Qualitatively speaking, the conversion residue V_{res} (i.e., the conversion error in the first stage) can be considered as having two components

$$V_{res} = E_q + E_{slope} \tag{1}$$

where E_q represents the quantization error, which is the same as in the DT-SAR. E_{slope} represents the additional error caused by the input variation, which is unique in the CT-SAR. E_{slope} happens when the CT input crosses a particular decision threshold and moves in the opposite direction with the subsequent binary search. Assuming the DAC has no redundancy, and such threshold crossing happens at the $(N-k+1)$th comparison (N is the total number of SAR comparisons), E_{slope} can be approximated by the input signal variation from the $(N-k+1)$th comparison to the end of the LSB comparison

$$E_{slope} \approx |\Delta V_{in}| \leq k \cdot T_{SAR} \cdot A \cdot 2\pi \cdot f_{in} \tag{2}$$

where T_{SAR} represents a single SAR cycle time, A and f_{in} are the input signal amplitude and frequency, respectively. The worst case E_{slope} happens at the MSB decision with $k = N$, as V_{in} has the longest time to drift away from the critical decision threshold. To minimize E_{slope}, it is preferred to reduce N, but this would reduce the CT-SAR resolution, leading to increased E_q. An effective way to reduce E_{slope} is to reduce T_{SAR}, as shown in Fig. 4(c). To this end, dynamic logic [2], [5] and asynchronous clocking [16] can be used. CMOS scaling also helps as it naturally decreases T_{SAR}.

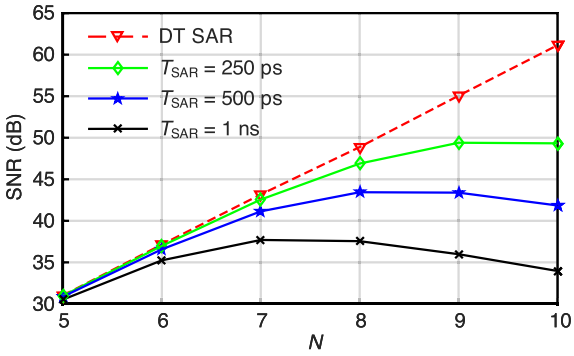


Fig. 5. SNR versus T_{SAR} and number of conversion cycles (N).

Fig. 5 shows the plot of simulated SNR versus N for the CT-SAR, assuming a full-swing 1-MHz input. Here, the SNR is defined as the signal power divided by the power of the conversion residue V_{res} at the end of the LSB comparison. When N is small, SNR is limited by E_q. As N increases, E_q decreases exponentially, and the SNR starts to be limited by E_{slope}. When N is very large, the SNR decreases because E_{slope} increases with N as indicated in (2). Shortening T_{SAR} can reduce E_{slope} and increase SNR, but it increases the design complexity and power consumption. Finally, the technology eventually places a lower bound for T_{SAR}. For example, in the 40-nm CMOS process, T_{SAR} is limited to about 200 ps. A more advanced process is required to further reduce T_{SAR}.

To further suppress E_{slope}, redundancy can be added in the SAR DAC [25]–[27]. The intuition is that the input signal

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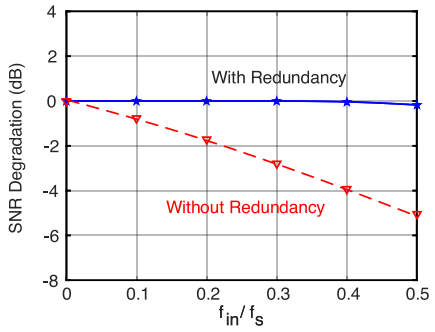


Fig. 6. Comparison with SNR degradation with/without redundancy.

variation can be considered as incorrect conversion results in the prior MSB decisions. Hence, as long as the added redundancy is larger than E_{slope} specified in (2), the DAC output can still catch up and track the time-varying input, as shown in Fig. 4(d).

As shown in (2), E_{slope} is larger for the MSB bits and smaller for the LSB bits. Thus, more redundancy should be allocated for the MSB bit ($k = N$), while less redundancy is needed for the LSB bit ($k = 1$). The optimal way to arrange the redundancy should follow (2). Thus, we can derive that the CDAC bit weight $\{W_k\}$ should be assigned in the following way to support a Nyquist-rate input:

$$W_k = \begin{cases} 1, & \text{for } k = 1 \\ \sum_{i=1}^{k-1} W_i - 2^{B-1} \cdot k \cdot T_{\text{SAR}} \cdot 2\pi \cdot f_{\text{in,Nyq}}, & \text{for } k > 1 \end{cases} \tag{3}$$

where B represents the effective number of bit of the CDAC. To simplify real implementation, the actual weight value can be rounded to its nearest integer.

To visualize the benefit of redundancy, Fig. 6 shows the plot of SNR degradation versus input signal frequency assuming $T_{\text{SAR}} = 500$ ps, the ADC sampling rate $f_s = 2$ MHz, and $B = 7$. The SNR degradation refers to the SNR difference between the CT-SAR and its corresponding DT-SAR. Without redundancy, there is appreciable SNR degradation as the input signal frequency increases. By contrast, with redundancy embedded, almost no SNR degradation is achieved over the entire Nyquist band, clearly demonstrating the benefit of adding redundancy.

This article uses both T_{SAR} minimization techniques and redundancy to reduce the CT-SAR error, so that V_{res} can be made comparable to that of a DT-SAR. As a result, this article obviates the need for the analog prediction filter or LC lattice filter, reducing chip area, power, and design complexity.

C. Inter-Stage Amplifier Operation With a Time-Varying V_{res}

As shown in Fig. 3, another key difference between the proposed ADC and the conventional DT two-step ADC is that the inter-stage amplifier in this article observes a continuously varying residue rather than a fixed dc-like residue. Thus, one question may arise that whether the inter-stage amplifier can work properly with a time-varying V_{res}, especially when a DA is adopted to reduce power and noise.

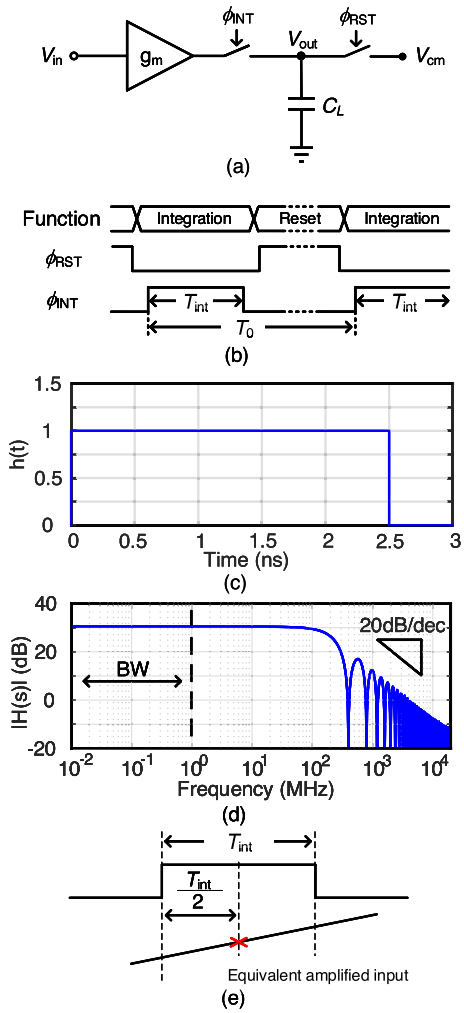


Fig. 7. (a) DA schematic. (b) Timing diagram. (c) Window function $h(t)$. (d) DA response $H(\omega)$. (e) Equivalent time-domain amplified signal point.

To analyze the operation of a DA in the presence of a time-varying input, let us consider a simplified circuit model as shown in Fig. 7(a). During the reset phase, the output C_L is short to a CM voltage. During the amplification phase, the DA works as a transconductor g_m that integrates the input onto C_L with the fixed time window T_{int}. Mathematically speaking, the DA output V_{out} is given by

$$\begin{aligned} V_{\text{out}}[n] &= \int_{nT_0}^{nT_0+T_{\text{int}}} \frac{g_m \cdot V_{\text{in}}(t)}{C_L} \cdot dt \\ &= \frac{g_m}{C_L} \cdot V_{\text{in}}(t) \circledast h(t)|_{t=nT_0+T_{\text{int}}} \end{aligned} \tag{4}$$

where $V_{\text{out}}(n)$ is the output after the nth integration, T_0 is the total time period consisting of both the reset and the integration phases [see Fig. 7(b)]. As shown in the right-hand side of (4), this integration process is equivalent to a convolution with a window function $h(t)$ [see Fig. 7(c)], and then sampled at $t = nT_0$. Thus, the overall transfer function of DA is equivalent

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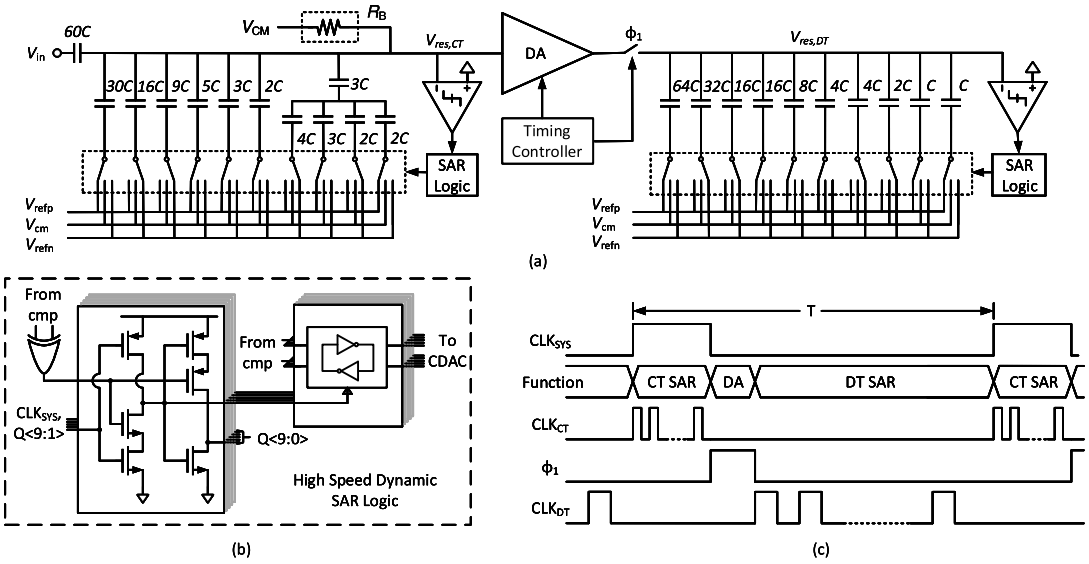


Fig. 8. Proposed CT-SAR-assisted two-step ADC. (a) Top-level schematic. (b) Dynamic SAR logic. (c) Timing diagram.

to a sinc function

$$H(\omega) = \frac{g_m \cdot T_{\text{int}}}{C_L} \cdot \text{sinc}(\omega \cdot \frac{T_{\text{int}}}{2}) \cdot e^{j\omega \cdot \frac{T_{\text{int}}}{2}}. \quad (5)$$

Fig. 7(d) shows an example magnitude response with $T_{\text{int}} = 2.5$ ns and a nominal dc gain of 30 dB. Within the signal bandwidth (BW) of 1 MHz, the DA works like a normal amplifier with a fixed gain. At out-of-band high frequencies, the DA has a low-pass response due to its integration behavior. As a result, it can provide an inherent mild first-order anti-aliasing capability. This low-pass response also filters out the wide-band thermal noise before the DA, leading to the significantly reduced sampling kT/C noise from the first stage. Fig. 7(e) also shows a simplified time domain view of the DA with a ramp-like input. The DA is equivalent to amplify a sampled middle point (or the time-average point) of the input. From this viewpoint, we can see that the DA operation with a time-varying input can be mapped to a sampled input case.

III. Prototype ADC Implementation

Fig. 8(a) shows the schematic of the prototype 2-MS/s ADC designed in 40-nm CMOS process. For simplicity, only the single-ended configuration is shown, but the real implementation is fully differential. In the first stage, the target effective number of bits B is chosen to be 7 to minimize the amplitude of V_{res} and relax the linearity requirement of the inter-stage amplifier. Its total number of comparisons N is set to 10 to allocate sufficient redundancy to tolerate input signal variation and capacitor mismatch. The unit capacitor C is 1 fF. A bridge capacitor is used to implement lower LSB capacitors without using even smaller unit capacitors. The input capacitor and the total CDAC capacitor are set to 60 fF and 68 fF, respectively, to balance the tradeoff among the ADC input signal swing, the attenuation factor, the chip area, and the ADC input and reference buffer requirement. The raw matching of these small capacitors cannot meet the requirement of the target 13-bit resolution. Thus, a one-time foreground capacitor

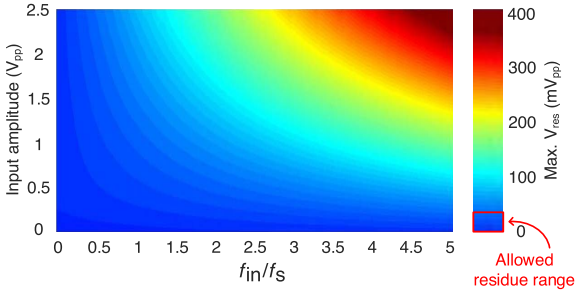


Fig. 9. Maximum residue voltage max(V_{res}) as a function of input signal frequency and amplitude.

mismatch calibration is applied as in [16], [28]. The size of the pseudo-resistor R_B is chosen such that the high-pass corner frequency is around 50 Hz.

Considering the 40-nm process, design complexity, and power consumption, T_{SAR} is set to 500 ps. The dynamic logic used to shorten T_{SAR} is shown in Fig. 8(b) [18]. Fig. 8(c) shows the ADC timing diagram. The first-stage CT-SAR is triggered at the rising edge of the system clock CLK_{SYS} and runs asynchronously for ten cycles. In total, the first-stage SAR takes about 5 ns to finish. Even though this is much faster than the Nyquist rate, its power increase is rather mild. In the 40-nm CMOS process, to meet this speed requirement, the transistor sizes for the logic gate and the comparator can still be kept small. As a result, the increase in the comparator power and SAR logic power are insignificant. The DA is triggered by the falling edge of CLK_{SYS} and it takes up 2.5 ns to achieve a gain of 32. Note that T_{SAR} may change due to process, voltage, and temperature (PVT) variation. If there is a large time difference between the end of first-stage SAR cycles and the DA start time (i.e., the falling edge of CLK_{SYS}), it would result in increased first-stage conversion residue amplitude. To address this issue, T_{SAR} is adjusted via tunable delay cells in the foreground to minimize this time difference.

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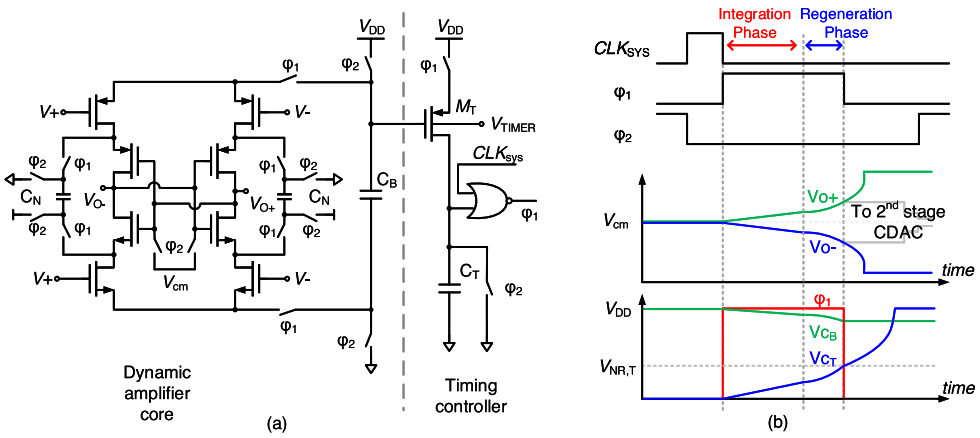


Fig. 10. Proposed FIB-DA. (a) Schematic. (b) Timing diagram and waveforms for key circuit nodes.

With the aforementioned ADC configuration, Fig. 9 shows the simulated amplitude of the equivalent amplified $V_{\rm res}$ by the DA as a function of both the input signal frequency and amplitude. The prototype ADC can support an input signal frequency covering the whole Nyquist BW with a peak-to-peak signal swing of 2.5 V without an appreciable increase in $V_{\rm res}$. For a smaller input amplitude of 20% full swing, the proposed CT-SAR can tolerate up to 2.5 f_s of input signal frequency. $V_{\rm res}$ increases when both input amplitude and frequency are large, but this situation can be prevented by a low-pass anti-aliasing filter before the ADC. Ensuring a small $V_{\rm res}$ of around 20 mV peak-to-peak significantly relaxes the linearity requirement of the inter-stage amplifier, which permits the use of an open-loop DA to replace the conventional closed-loop static amplifier to reduce power and noise.

The schematic of the proposed FIB-DA is shown in Fig. 10(a). It uses a inverter-based CMOS input stage to double the transconductance compared to a conventional DA with a single nMOS or pMOS input pair [7], [19]–[21]. A cross-coupled inverter is inserted at the output to provide positive feedback to boost the DA voltage gain [29], [30]. The amplifier is powered by a 3.2-pF battery capacitor C_B, which is re-charged to $V_{\rm DD}$ and GND during the DA reset phase. The battery capacitor C_B isolates the DA operation from the power supplies $V_{\rm DD}$ and GND during the amplification phase, and thus, provides stronger rejection to the input CM voltage variations. Moreover, unlike a conventional DA whose output CM voltage is typically not well controlled and sensitive to PVT variations, the proposed FIB DA ensures a constant output CM voltage, owing to the use of the floating battery capacitor. It inherently guarantees that the nMOS pair current must match the pMOS pair current, and thus, the output CM current has to be zero. This obviates the need for an explicit output CM feedback loop [31]. Having a stable output CM voltage allows a wide output signal swing and a large amplifier gain. C_N of 30 fF serves as the internal integration capacitor to reduce the DA BW and its input-referred noise. In a conventional DA, each nMOS or pMOS pair would have its own integration capacitor. By contrast, this article shares a single C_N between nMOS and pMOS pairs. This cuts down the capacitor size by four times (i.e., a single C_N rather than two $2C_N$s). In the reset phase, C_N is connected to $V_{\rm DD}$ and GND.

The timing diagram and waveforms for key circuit nodes are plotted in Fig. 10(b). At the falling edge of $\rm CLK_{SYS}$, ϕ_1 goes high and the DA is turned on. The DA output voltages V_{o+} and V_{o-} start to depart from each other due to DA input integration. After a certain time, the voltage across C_N becomes high enough to enable the cross-coupled inverter, which then leads to the exponential growth of V_{o+} and V_{o-} due to the positive feedback. Note that although an output differential mode voltage is developed, the output CM voltage remains nearly unchanged, which is ensured by using the battery capacitor C_B to power the DA, as explained earlier. As time goes by, C_B loses charge and its top-plate voltage V_{C_B} starts to drop below $V_{\rm DD}$. As a result, the transistor M_T starts to turn on and charge up C_T. Once the C_T voltage V_{C_T} reaches the logical threshold of the NOR gate, ϕ_1 goes low, which ends the amplification phase. Note that the DA amplification time depends on how fast M_T charges C_T, and thus, can change with PVT variations. In this article, to keep the DA gain constant, the DA time is off-chip foreground calibrated by tuning the back-gate voltage of M_T using $V_{\rm TIMER}$ as in [29]. Background calibration can also be done as in [30].

Unlike the simple DA in Section II-C that always works in the linear integration phase, the proposed FIB DA has two operation phases: a linear integration phase and a positive-feedback regeneration phase. The time of the integration phase and the DA BW are primarily set by the value of the integration capacitor C_N. The total DA operation time (including both operation phases) is set by C_B, M_T, and C_T. Adjusting them mainly changes the time that the DA spends in the regeneration phase. More detailed discussions can be also found in [30] and [32].

Also different from the simple DA in Section II-C, the corresponding $h(t)$ of the proposed FIB DA is not constant but time-varying, as shown in Fig. 11(a). Nevertheless, the slight time dependence of $h(t)$ only mildly changes the equivalent transfer function of the proposed DA. Comparing to Fig. 11(b) and Fig. 7(d), the only difference is the removal of deep notches. The in-band flatness and the out-of-band -20 dB/dec-low-pass behavior are maintained.

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TABLE I
PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART ADCS

Specifications	[1]	[4]	[5]	[6]	[14]	[15]	**This work**
Architecture	SAR	SAR	Pipeline	Pipe-SAR	CT pipeline	CT pipeline	**CT two-step SAR**
kT/C noise attenuated	✗	✗	✗	✗	✓	✓	✓
1st-stage input capacitance (pF)	9	16	3.2	4	Resistive	Resistive	**0.12**
Area (mm^2)	0.18	0.32	1	0.054	1.9	5.1	**0.01**
Technology	40nm	40nm	180nm	65nm	180nm	28nm	**40nm LP**
Resolution (bits)	14	15	15	13	11	N/A	**13**
f_s (MS/s)	0.032	0.02	10	50	26	9000	**2**
Input Swing ($V_{pp,diff}$)	1.6	1.8	2.5	2.4	3.6	N/A	**2.5**
$SFDR_{NYQ}$ (dB)	78.5	95.1	95.4	84.6	67	73	**79.4**
$SNDR_{NYQ}$ (dB)	69.7	74.1	76.8	70.9	61.1	68	**71.7**
$ENOB_{NYQ}$ (bit)	11.3	12	12.5	11.5	9.9	11	**11.6**
Total Power (uW)	0.352	1.17	5100	1000	26700	2330000	**25.2**
FoM_w (fJ/step)	4.4	14.1	45	6.9	1113	495	**3.9**
FoM_s (dB)	176.3	173.4	170	174.9	148	160	**177.8**

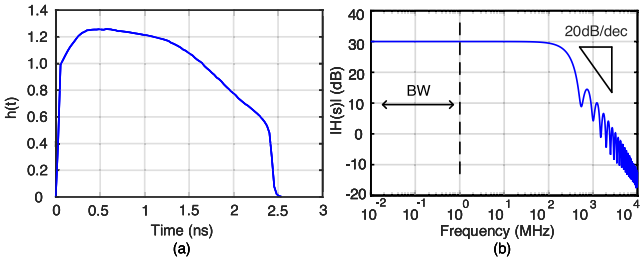


Fig. 11. Proposed FIB-DA. (a) Simulated $g_m(t)$. (b) Frequency response.

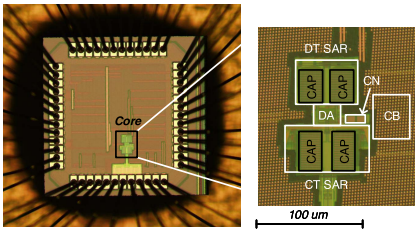


Fig. 12. Chip photograph.

IV. MEASUREMENT RESULTS

Fig. 12 shows the die photo of the prototype ADC in a 40-nm low-power (LP) CMOS process. The ADC core area is 0.01 mm^2. The supply voltage used by the first-stage CT-SAR and the DA is set to 1.1 V to enhance the operation speed and support a wide input signal swing. The second-stage DT-SAR uses a supply voltage of 0.7 V to reduce power consumption. At the sampling rate of 2 MS/s, the ADC consumes in total 25.2 μW of power, where the first-stage CT-SAR, the DA, and the second-stage DT-SAR consume 7.1, 12.2, and 5.9 μW, respectively. Inside the first-stage CT-SAR, the comparator, the digital circuits, and the DAC consume 2.4, 3, and 1.7 μW, respectively. Inside the second-stage DT-SAR, the comparator, the digital circuits, and the DAC consume 1.9, 1.9, and 2.1 μW, respectively.

Fig. 13(a) and (b) shows the measured spectrum with a low-frequency signal and a near Nyquist-rate input signal, respectively. With a full-swing input at 100 kHz, the measured SNDR and spurious-free dynamic range (SFDR) are 73.5 and 87.8 dB, respectively. With a 0-dBFS near-Nyquist-rate input of 950 kHz, the measured SNDR and SFDR are

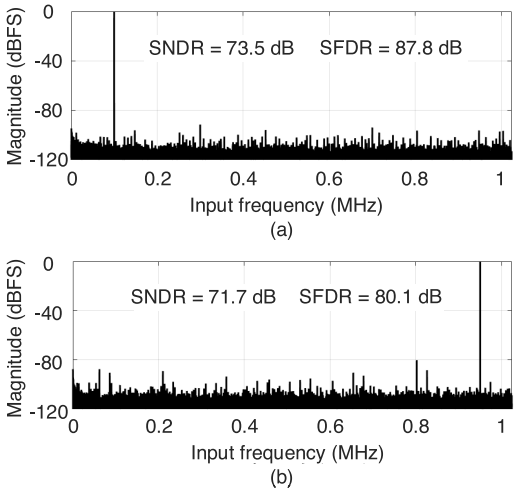


Fig. 13. Measured spectrum with (a) 100- and (b) 950-kHz input.

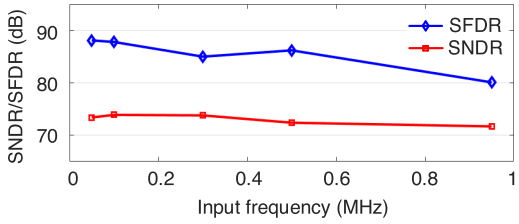


Fig. 14. Measured SNDR and SFDR versus input frequency.

71.7 and 80.1 dB, respectively. As mentioned in Section III, the foreground calibration is performed for both the first-stage and the second-stage capacitor mismatches, without which the SNDR and SFDR would be limited to 60 and 65.4 dB, respectively. The gain and the offset of the DA, and the offset of the first-stage comparator are also calibrated. In addition, the first-stage SAR logic delay is calibrated to minimize the time difference between the end of first-stage SAR cycles and the DA starting edge.

Fig. 14 shows the measured SNDR and SFDR versus the input frequency. Fig. 15 shows the input amplitude sweep. The measured dynamic range is 73.6 dB. These measurement results show that the prototype works properly as a

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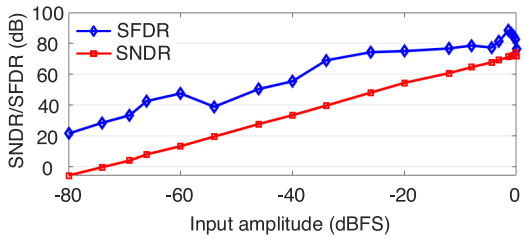


Fig. 15. Measured SNDR and SFDR versus input amplitude.

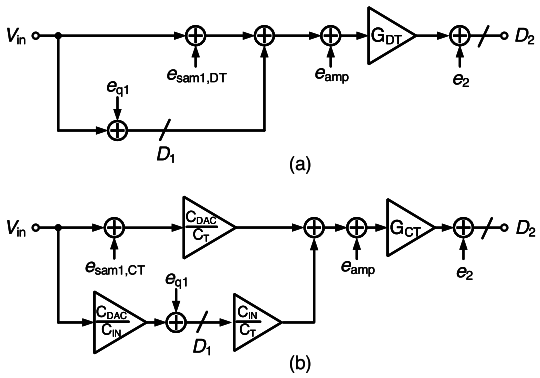


Fig. 16. Noise linear model for (a) conventional DT-SAR ADC and (b) proposed CT-SAR ADC.

Nyquist-rate ADC over various input amplitude and frequency settings.

Table I shows the performance summary and compares it with other state-of-the-arts. The input capacitor of this work is orders of magnitude smaller than others with similar SNDR, which is made possible by the CT front end with sampling noise suppression. Its chip area of 0.01 mm^2 is also significantly smaller than others, due to the reduced capacitor size. The Walden and Schreier figure-of-merits (FoMs) with the Nyquist frequency input are 3.9 fJ/conversion-step and 177.8 dB, respectively, and are in line with the state-of-the-arts.

V. Conclusion

This article presented a two-step ADC architecture with a first-stage CT-SAR. By removing the S/H circuit, the proposed ADC breaks the seemingly fundamental tradeoff between the input capacitor size and the sampling noise. The CT-SAR conversion error is minimized by adding redundancy and accelerating the SAR speed. With the substantial reduction in the input capacitor size, it is envisioned that the power, area, and design complexity of the ADC driver and the reference buffer can be relaxed, leading to significant benefits on the system level.

Appendix
Noise Analysis of the Proposed ADC

The noise models for the conventional DT-SAR ADC and the proposed CT-SAR ADC are shown in Fig. 16(a) and (b), respectively. For simplicity of purpose, we only consider the first-stage sampling noise e_{sam1}, the inter-stage amplifier noise e_{amp}, and the second-stage noise e_2 (including the second-stage sampling noise, the quantization noise, and the comparator noise).

For the DT-SAR ADC shown in Fig. 16(a), the total input-referred noise $e_{\text{tot,DT}}$ can be derived as

$$\begin{aligned} e_{\text{tot,DT}} &= e_{\text{sam1,DT}} + e_{\text{amp}} + \frac{e_2}{G_{\text{DT}}} \\ &= \frac{kT}{C_{\text{DAC}}} + e_{\text{amp}} + \frac{e_2}{G_{\text{DT}}} \end{aligned} \tag{6}$$

where G_{DT} represents the inter-stage gain in the DT case and C_{DAC} represents the total CDAC capacitance.

For the CT-SAR ADC shown in Fig. 16(b), the total input-referred noise $e_{\text{tot,CT}}$ can be derived as

$$\begin{aligned} e_{\text{tot,CT}} &= e_{\text{sam1,CT}} + \frac{C_T}{C_{\text{DAC}}} \cdot \left(e_{\text{amp}} + \frac{e_2}{G_{\text{CT}}} \right) \\ &= 4kT R_{\text{eq}} \cdot \text{BW}_{\text{DA}} + \frac{C_T}{C_{\text{DAC}}} \cdot \left(e_{\text{amp}} + \frac{e_2}{G_{\text{CT}}} \right) \end{aligned} \tag{7}$$

where C_T represents the total capacitance at the first-stage comparator input node, G_{CT} represents the inter-stage gain in the CT case, BW_{DA} represents the DA BW, and R_{eq} represents the equivalent resistance of the first-stage CT input sampling network.

For a fair comparison, we assume that both the CT and DT ADCs have the same nominal resolution (i.e., the quantization noise). Thus, it is easy to derive that $G_{\text{DT}} = G_{\text{CT}} \cdot (C_T / C_{\text{DAC}})$. Plugging it in (7) and rearranging (7), we have

$$e_{\text{tot,CT}} = \frac{\text{BW}_{\text{DA}}}{1/(R_{\text{eq}} C_{\text{DAC}})} \cdot \frac{kT}{C_{\text{DAC}}} + \frac{C_T}{C_{\text{DAC}}} \cdot e_{\text{amp}} + \frac{e_2}{G_{\text{DT}}}. \tag{8}$$

Comparing (6) and (8), there are two differences. First, the first-stage sampling noise is greatly attenuated in the proposed CT-SAR topology. In the conventional DT-SAR ADC, with the noise power spectral density (PSD) being $4kTR_{\text{eq}}$ and the noise BW being $1/(R_{\text{eq}}C_{\text{DAC}})$, the first-stage sampling noise is kT/C_{DAC}. By contrast, in the CT-SAR ADC, the sampling noise PSD and BW are de-coupled. The effective noise BW is not $1/(R_{\text{eq}}C_{\text{DAC}})$, but BW_{DA}. The sampling noise reduction ratio is $\text{BW}_{\text{DA}}/[1/(R_{\text{eq}}C_{\text{DAC}})]$, which is the ratio of the DA BW over the first-stage sampling network BW. By having a small C_{DAC} (e.g., 60 fF) and a small R_{eq}, the first-stage sampling network BW can be made much larger than the DA BW, leading to the first-stage sampling noise much smaller than kT/C_{DAC}. Second, the CT operation with the capacitive input network comes with a penalty of input signal attenuation, which results in the increased inter-stage amplifier noise by the factor of C_T/C_{DAC}. To avoid high-power consumption of the inter-stage amplifier, the overall ADC noise budgeting is optimized in this design. Since the first-stage sampling noise is reduced, a larger portion of the noise budget can be assigned to the inter-stage amplifier, which lowers its power consumption. Overall, the proposed CT-SAR technique can be used to reduce the first-stage capacitor sizes and the core ADC area without causing a large sampling noise penalty. Moreover, with the significantly reduced first-stage capacitor sizes, the performance requirements of the ADC driver and the reference buffer can be relaxed, which can lead to the power saving on the system level.

Table II shows the input-referred noise breakdown of the prototype ADC. The measurement result matches well with the simulation result.

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TABLE II
NOISE BUDGETING OF THE PROPOSED CT-SAR ADC

	Simulation results	Measurement results
Quantization noise	76.8 μV	N/A
DA noise	171.1 μV	N/A
1st-stage noise	82.8 μV	N/A
2nd-stage noise	42.5 μV	N/A
Total noise	209.4 μV	215.7 μV

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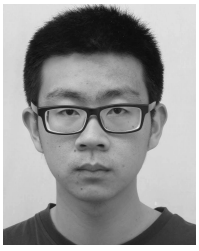
Mr. Shen was a recipient of the IEEE Solid-State Circuits Society Predoctoral Achievement Award in 2019, the Graduate Continuing Fellowship from UT Austin in 2019, the Samsung Fellowship in 2011, and the National Scholarship in 2012.

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